



BEML LIMITED BANGALORE

DRCA NO.	RS(3R)/CO/P/B740/ A1
DOC. NO.	GR/TD/3328
DATE	13.12.2017
REV. No.	01
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Vehicle Control Circuits Description & Drawings

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1. Introduction

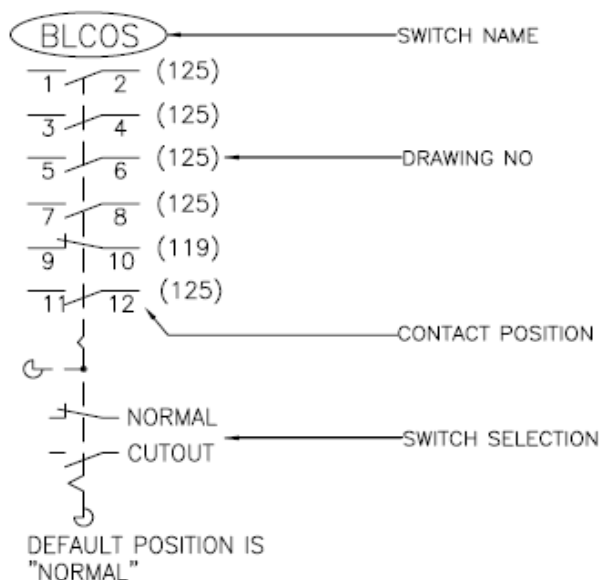
This chapter gives the preliminary description of Vehicle Control Circuit (VCC) drawings for KMRCL and provides technical information for the train operator/maintenance Personnel. The information includes train control circuit operating logic including various interlocking by relays, contactors, cam switch, MCB (Miniature Circuit Breaker) etc, applied on the KMRCL train.

The manual and VCC drawings are divided into 11 broad categories as follows;

- 0 General
- 1 Train
- 2 Vehicle Structure & interior Fitting
- 3 Bogie & Suspension
- 4 Gangway & Coupler
- 5 Traction System
- 6 Brake System
- 7 Auxiliary Electric System
- 8 Door System
- 9 Air Conditioning System
- 10 Train Integrated Management System
- 11 Communication System

This manual should be read in conjunction with VCC drawings.

Every VCC sheet show all the relays, contacts, rotary switches and push buttons, their pin allocations and drawing number on which their contact is used .



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Each VCC drawing has designation of each component, function, location, connected wire nos and reference circuit diagram sheet also. All abbreviations used on vehicle control circuits are listed alphabetically in every sheet.

2. Alphabetical Index

Abbreviation	Description
A	
AC1,2CB	Saloon Air-con1,2 Circuit Breaker
ACP	Auxiliary Control Panel for PA/PIS
ADCLp	All Doors Closed Indicator
ADCR	All Doors Closed Relay
ADMV	Auto Drain Magnetic Valve
ADU	Air Dryer Unit
AOFFS	Auxiliary Off Switch
APFLp	Auxiliary Power Fault Indicator Lamp
APSFR	Auxiliary Power Supply Fault Relay
APSCB1, 2	APS Circuit Breaker1, 2
ASDR	Auxiliary Shut Down Relay
ATPCB	ATP Circuit Breaker
AUX ON	Auxiliary On Switch
B	
BBRSCB	BBRS Circuit Breaker
BCB	Battery Control Box
BCBCB	Battery Control Box Circuit Breaker
BCCOS1, 2	Brake Cylinder Cut Out Switch1, 2
	(Bogie Isolation Cock: BIC)
BCU	Brake Control Unit
BCPS	Brake Cylinder Pressure Switch
BECU	Brake Electronic Control Unit
BECUCB1, 2	BECU Circuit Breaker1, 2
BF1, 2	Battery Fuse1, 2
BIC	Battery Isolation Contactor
BIS	Battery Isolation Switch
BLCB	Brake Loop Circuit Breaker
BLPR	Brake Loop Proving Relay
BPIMV	Brake Pipe Pressure Isolation Magnetic Valve
BPPS	Brake Pipe Pressure Switch
BUVDR	Battery Under Voltage Detection Relay

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C	
CACCB	Cab A/C Circuit Breaker
CCCB	Coupling Control Circuit Breaker
CC1,2CB	Controlling Cab Circuit Breaker
CCTVCB	CCTV Circuit Breaker
CM	Compressor Motor
CMCB	Compressor Motor Circuit Breaker
CMLCB	Cab Main Light Circuit Breaker
COLR	Call-On Light Relay
COLS	Call-On Light Switch
CSDOS (L)/(R)	Cab Side Door Open Limit Switch (Left)/(Right)
CSPDS	Cab Saloon Partition Door Limit Switch
CC	Carborne Controller
D	
DCAPB (L)	Door Close Announcement Push Button, Left
DCAPB (R)	Door Close Announcement Push Button, Right
DCRCB	Door Close Circuit Breaker(Right)
DCLCB	Door Close Circuit Breaker(Left)
DPLCB	Door Proving Loop Circuit Breaker
DCLPB1, 2	Door Close Push Button (Left)1, 2
DCLR	Door Close Relay(Left)
DCRPB1, 2	Door Close Push Button (Right)1, 2
DCRR	Door Close Relay (Right)
DMS	Door Mode Switch
DOLPB11, 12, 2	Door Open Push Button, Left 11, 12, 2
DORPB11, 12, 2	Door Open Push Button, Right 11, 12, 2
DOLR	Door Open Relay (Left)
DORR	Door Open Relay (Right)
DPLCB	Door Proving Loop Circuit Breaker
DPR	Door Proving Relay
DSDBS	Deadman Safety Device Bypass Switch
DCS	Data Communication Modem
E	
EB(1,2,3,4)	Earth Brush (1,2,3,4)
EBLR 1(2)	Emergency Brake Loop Relay
EBMV	Emergency Brake Magnetic Valve
EDB	Electrical Distribution Board
EBLp	Emergency Brake Applied Indicator
EBLRCB	Emergency Brake Loop Relay Circuit Braker
EBMVCB	Emergency Brake Magnet Valve Circuit Breaker

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EBVR	Emergency Brake Valve Relay
EDCU	Electronic Door Control Unit
ELCB1, 2,3,4	Emergency Light Circuit Breaker1, 2,3,4
ESKCB	ESK Circuit Breaker
EOSS	Electro Optical Speed Sensor
F	
FL	Flasher Light
FR	Fault Relay in VAC
FSB	Full Service Brake
G	
GWL1, 2	Gangway Light1, 2
H	
HBTLp	HSCB Trip Indicator
HCR	Head Control Relay
HL(L/R)	Head Light (Left)/(Right)
HLLCB	Head Light Left Circuit Breaker
HLRCB	Head Light Right Circuit Breaker
HMV1	Horn Magnetic Valve (Low)
HMV2	Horn Magnetic Valve (High)
HSCB	High Speed Circuit Breaker
HTEB	High Tension Earth Box
K	
KOR	Key On Relay
L	
LCAR	Last Cab Activated Relay
LDICB	Local Door Interlock Circuit Breaker
LDRCB	Local Door Right Circuit Breaker
LDLCB	Local Door Left Circuit Breaker
LTEB	Low Tension Earth Box
LTPB	Lamp Test Push Button
LTR	Lamp Test Relay
LVLp	Line Voltage Indicator
M	
MCP	Master Control Panel for PA/PIS
MCCMV	Mechanical Coupler Coupled Magnetic Valve
MCUCMV	Mechanical Coupler Uncoupled Magnetic Valve

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MS	Mode Selector
MSOCB	Maintenance Socket Outlet Circuit Breaker
O	
OCOLp	Other Cab Occupied Indicator
P	
PAACB	PA Amplifier Circuit Breaker
PBLp	Parking Brake Applied Indicator
PBCB	Parking Brake Circuit Breaker
PBR	Parking Brake Relay
PBMV	Parking Brake Magnetic Valve
PIBCB	Passenger Information Board Circuit Breaker
PE	Protective Earthing
R	
RP	Relay Panel
S	
SAC1,2CCB	Saloon Air-con 1, 2 Control Circuit Breaker
SCS	Safety Cut Out Switch
SCSR	Safety Cut Out Switch Relay
SIVKR	SIV Contactor Relay
SOCB	Socket Outlet Circuit Breaker
SSCB	Shore Supply Circuit Breaker
SSICB	System Status Indicator Circuit Breaker
SSKR	Shore Supply Contactor Relay
STUR	Start-Up Relay
SUCB	Start-Up Circuit Breaker
T	
TBC	Traction Brake Controller
TBCCB	Traction Brake Controller Circuit Breaker
TCPU1, 2CB	TCPU1, 2 Circuit Breaker
TCR	Tail Control Relay
TDOCB	TCMS Digital Output Circuit Breaker
TL(L)/(R)	Tail Light (Left)/(Right)
TLCB	Train Line Circuit Breaker
TLFR	Tail Light Flasher Relay
TLSC1R	Train Line Supply Contactor 1 Relay
TRUCB	Train Radio Unit Circuit Breaker
TCMS	Train Control Management

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V	
VDUCB	Video Display Unit Circuit Breaker
VFCB1, 2	VVVF Inverter1, 2 Circuit Breaker
VVVFFLp	VVVF Fault Indicator
VVVFFR	VVVF Fault Relay
VAC	Ventilation and Air Conditioning
U	
UCCB	Uncoupling & coupling Circuit Breaker
UCPB	Uncoupling Pushbutton
V	
VDU	Video Display Unit
W	
WWCB	Windscreen Wiper Circuit Breaker
Z	
ZVAR	Zero Velocity Auxiliary Relay

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3. General

3.1 Introduction

The general part drawings of the vehicle control circuits provide useful information such as Drawing List, Wiring numbers, description, Train-lines, symbols etc. For a good grasp of the circuit itself, the general part drawings must be referred to.

Detail information of the general part drawings is as below:

Description	Dwg No
Drawing List, Vehicle Control Circuit	942-58099
Classification	942-58100
Wire Numbers, Description	942-58101
Symbols	942-58102
Train-Lines, Control	942-58103
Train-Lines, Screen Shield	942-58104
Train-Lines, Low Tension Power	942-58105
Train-Lines, High Tension Power	942-58106

3.2 Drawing List (942-58099)

The list of the vehicle control circuit is shown on the drawing, 942-58099. The drawings are listed according to their system and numbers. Through the drawing, one can know drawing list, its system and car type applied.

System Hierarchy				Car type application		
Drawing number				Drawing Title		
GENERAL						
NO.	DWG NO.	REV NO.	DESCRIPTION	CAR TYPE		
1	942-58099	—	DRAWING LIST	DMC	TC	MS
2	942-58100	—	CLASSIFICATION	○	○	○
3	942-58101	—	WIRING NUMBERS, DESCRIPTION	○	○	○
4	942-58102	—	SYMBOLS(1/4)	○	○	○
5	942-58102	—	SYMBOLS(2/4)	○	○	○
6	942-58102	—	SYMBOLS(3/4)	○	○	○
7	942-58102	—	SYMBOLS(4/4)	○	○	○
8	942-58103	—	TRAIN LINES, CONTROL(1/4)	○	○	○

3.3 Classification (942-58100)

The system classification of the vehicle control circuit is shown on the drawing, 942-58100. All the vehicle control circuits are classified according to their system hierarchy and the system of each drawing is shown on the top-right corner.

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3.4 Wiring Numbers Description (942-58101)

Drawing 942-58101, gives details of the wire numbering system.

3.4.1 Conductor Designation

Followings are description of conductors designated:

- HV: Main Circuit 750V
- ED: Propulsion Circuits supply of AC traction Motors.
- AP: Auxiliary power cables for 3Ph415V and 230VAC, 50Hz.
- BA: Conductors directly supplied by the battery (110VDC)
- S: Shielded cables for measuring and analogue voltage signals Data communication link Cable connected to special equipment for signals such as radio antenna

3.4.2 Numbering

Cables have their own number that is composed of 5 digits which stand for its characteristics such as car type, system, AC or DC line, etc. Meaning of each digit of wire numbers is as follow:

(a) First digit

It represents the car type in which the cable used.

If the digit is:

- None: the cable is train line that runs through 2 units
- D: The cable is used in DMC
- T: The cable is used in TC
- M: The cable is used in MC

(b) Second digit

It represents the cable no based on the System Hierarchy.

3.5 Symbol Description (942-58102)

All the symbols used on vehicle control circuits are summarized on the drawing.

3.6 Train Lines, Control (942-58103)

The drawings show all the train-lines for control. In the drawings, wire numbers, function, jumper cable pin numbers, drawing numbers on which shown the wire, etc. are incorporated.

3.7 Train Lines, Screen Shield (942-58104)

The drawings show all the train-lines for screen shield. In the drawings, wire numbers, function, jumper cable pin numbers, drawing numbers on which shown the wire, etc. are incorporated.

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3.8 Train Lines, Low Tension Power (942-58105)

The drawings show all the train-lines for low-tension power. In the drawings, wire numbers, function, jumper cable pin numbers, drawing numbers on which shown the wire, etc. are incorporated.

3.9 Train Lines, High Tension Power (942-58106)

The drawings show all the train-lines for high-tension power. In the drawings, wire numbers, function, jumper cable pin numbers, drawing numbers on which shown the wire, etc. are incorporated.

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4. Train Control Circuits

4.1 Introduction

In this section, circuit descriptions for Controlling Cab, start-up, System status indication and Train line supply Contractor circuits are provided as follows;

Description	Dwg No
Controlling Cab	942-58107
Start-Up Relay	942-58108
System Status Indication	942-58109
MCB Trip State Monitoring	942-58110
DC Train Line Supply Contactor	942-58111

4.2 Controlling Cab (942-58107)

One train set consists of two Driving Cabs. Controlling-cab circuit distinguishes between occupied cab and non-occupied cab.

The train can be controlled only from one cab; therefore the control circuit must be operational on the controlling-cab. The main controls of the train are Master controller (Mode Selector and Traction Brake Controller) with a dead-man safety device.

For the operation of train, FIRST, the driver should insert the master key into the key switch and turn it on (Mode selector is in OFF position and Traction Brake Controller is in BMAX position when the key is OFF).

4.2.1 Activation of HCR

When Mode Selector (MS) is moved to any position "REVERSE or WASH/ COUPLING or STANDBY or RESTRICTED MANUAL or ATP or ATO" in the occupied cab, the train control is activated.

In this state the occupied cab relays of HCRs and LCARs are energized.

Head Control Relays and Last Cab Activated Relay can deal with train control command, for example powering/braking, headlight on/off, Train moving direction, door open/close, emergency brake and etc.

4.2.2 Activation of TCR

In the rear cab of the train set, Tail Control Relay (TCR) will get energized at the same time of activation of HCR.

Tail Control Relays (TCR) of the occupied cab is to remain de-energized, as the normal close contact of HCRs are open and prevents train control from the rear cab.

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4.2.3 Activation of LCAR

The function of LCAR is of a memorized relay in which this occupied cab has been occupied before. The interlocking circuit of ASDR, HCR and TCR contacts energizes Last Cab Activated Relay (LCAR) of occupied cab.

The condition of HCR, TCR & LCARs in accordance with MS position is shown as below;

Key position	Leading DMC car			Rear DMC car		
	HCR1~6	TCR1,2	LCAR1,2,3	HCR1~6	TCR1,2	LCAR1,2,3
OFF	De-energized	De-energized	De-energized	De-energized	De-energized	De-energized
ON	Energized	De-energized	Energized	De-energized	Energized	De-energized

4.3 Start-up / Shut-down (942-58108)

4.3.1 Start-up

For the initialization / shutdown of the train system, start-up/shut-down control arrangements are provided. Detailed start up/shutdown route is shown in the below.

Pressing and releasing of the Auxiliary On (AUX ON) causes the train system to be powered-up according to the sequential control arrangement.

When the AUX ON button on the leading cab is pressed and released, each unit of train set gets to activate.

The train start up sequence is as below;

- Check the Current Collector position.
- Pressing and releasing of AUX ON.
- Operation of VVVF Inverter and APS.

4.3.2 Shut-down

When the AOFFS button is pressed and released in the occupied cab, VVVF Inverter and APS will shut-down.

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4.3.3 Reset

In the event of fault detection in the VVVF Inverter and APS, Both systems can be reset by pressing the reset button.

4.3.4 Functions

4.3.4.1 Start-up Relay (STUR)

A Start-up Relay (STUR) on each car is a momentary monitoring relay to give a starting signal to the control circuit of VVVF Inverter, APS and Battery control.

4.3.4.2 Auxiliary Shut-down Relay (ASDR)

When the AOFFS button is pressed and released, an Auxiliary Shut-down Relay (ASDR) is energized to give a signal of total shut-down to control circuit of VVVF Inverter APS and Battery control.

4.3.4.3 Auxiliary On (AUX ON)

This push button is to power up the train.

4.3.4.4 Auxiliary Off (AOFFS)

This push button is to shut down VVVF Inverter and APS powered equipment and the train will be totally powered off.

4.3.4.5 Reset (RESET)

In the event of some failure detection in VVVF / APS, the RESET push button is pressed and released to reset failure of VVVF Inverter / APS of that unit.

4.4 System Status Indication (942-58109)

(a) Other Cab Occupied Indicator (OCOLp)

OCOLp in the un-occupied cab would be illuminated when the TCR on the un-occupied DMC is energized.

(b) VVVF Inverter fault (VVVFFLp)

VVVFFLp in the activated driving cab would be illuminated when VVVF Inverter fault is detected on the each DMC/MC car.

(c) Line Voltage Indicator (LVLp)

LVLp in the activated driving cab would be illuminated when the traction supply is available at the current collector of the leading unit.

(d) HSCB Trip (HBTLp)

HBTLp in the activated driving cab would be illuminated when tripping of HSCB is detected on each DMC/MC car.

(e) Auxiliary Power Fault Indicator (APFLp)

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APFLp in the activated driving cab would be illuminated when Auxiliary Power System (APS) fault is detected on the each TC.

In DMC car, there are two APFLps. One installed on indicator panel of cab and other installed on the relay panel of DMC.

In TC car, APFLp is installed on the EDB panel.

(f) Cab VAC Fault Indicator (CVACFLp)

CVACFLp in the activated driving cab would be illuminated when any cab VAC unit fault is detected on the cab cars. There are two CVACFLp. One installed on operator panel #4 of cab and other installed on the relay panel of DMC.

(g) Saloon VAC Fault Indicator (SVACFLp)

SVACFLp in the activated driving cab would be illuminated when any passenger saloon VAC unit fault is detected on the either cars. There are two SVACFLp. One installed on operator panel #4 of cab and other installed on the relay panel of DMC.

(h) Parking Braking Applied Indicator (PBLp)

PBLp in the activated driving cab would be illuminated when PBR (Parking Brake Relay) is applied on the DMC. There are two PBLps. One installed on Indicator panel of cab and other installed on the relay panel of DMC.

(i) Emergency Brake Applied Indicator (EBLp)

EBLp in the activated driving cab would be illuminated when EBVR1 and EBVR2 (Emergency Brake Valve Relay) is de-energized.

(j) All Door Closed Indicator (ADCLp)

ADCLp in the activated driving cab would be illuminated when the ADCR (All Door Closed Relay) is energized on the leading cab. There are three ADCLps. The indicators installed on the Indicator panel, Operator panel #5 and Operator panel #6.

4.4.1 Lamp Test

Pressing of the "LTPB (Lamp Test push button)" in the driver's cab would carry out a self-test to ensure the correct operation of all indicators.

4.5 MCB Status Monitoring with TCMS (942-58110)

For the safety of train operation, TCMS in each car monitors following condition of all MCBs;

4.5.1 Monitoring of MCBs on the DMC

- ☐ MCBs for 110VDC on the MCB panel #1 is not tripped
- ☐ MCBs for 110VDC on the MCB panel #2 is not tripped

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- ☐ MCBs for ATP (CC) system is not tripped
- ☐ MCBs for AC group is not tripped
- ☐ MCBs for DC110V Interior light is not tripped
- ☐ MCBs for Brake group is not tripped
- ☐ MCBs for Door group is not tripped
- ☐ MCBs for VVVF Inverter group is not tripped

4.5.2 Monitoring of MCBs on the TC

- ☐ MCBs for Brake group is not tripped
- ☐ MCBs for Door group is not tripped
- ☐ MCBs for 110VDC group is not tripped
- ☐ MCBs for AC group is not tripped
- ☐ MCBs for DC 110V Interior light is not tripped
- ☐ MCBs for DC 110V Battery is not tripped

4.5.3 Monitoring of MCBs on the MC

- ☐ MCBs for Brake group is not tripped
- ☐ MCBs for Door group is not tripped
- ☐ MCBs for 110VDC group is not tripped
- ☐ MCBs for AC group is not tripped
- ☐ MCBs for DC 110V Interior light is not tripped
- ☐ MCBs for VVVF Inverter group is not tripped

4.6 DC Train line Supply Contactor (942-58111)

110V DC train line power is derived from D1, D2 line and supplied to train line via TLSC contact. The train lines negative have one point connection in occupied DMC.

4.6.1 Wire "D3"

The redundant DC 110V supply line (D3) is energized by TLSC2 via LCAR2. The load distribution from "D3" line is as follows;

- System Status Indication circuit
- Speed Control Circuit 1
- Emergency Brake Loop Circuit (Normal)
- Emergency Brake Magnet Valve Circuit
- Parking Brake Circuit
- Door Operation, Left Circuit
- Door Operation, Right Circuit

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- Door Proving Loop Circuit
- TCMS Circuit

4.6.2 Wire "D4"

The battery redundant line (D4) is energized by TLSC1 via LCAR1. The load distribution from "D4" line is as follows;

- Controlling cab circuit 2
- Speed control circuit 2
- Emergency Brake Loop Circuit (Redundant)

4.6.3 Train line "Ground Return Line (GRL)"

Main purpose of GRL is to make one point earthing/return connection through the train. All the DC negative feedback lines are connected to GRL1&2, which in turn is connected to earthing point of DMC car via TLSCs N/O contact.

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5. Vehicle Structure & Interior Fitting Circuits

5.1 Introduction

In this section, circuit descriptions for Lights, Windshield Wiper Circuit are provided.

	Description	Dwg No
.	Head & Cab Main Light	942-58112
.	Tail Light, FL & Console Light	942-58113
.	Saloon Lights, DMC	942-58114
.	Saloon Lights, TC/MC	942-58115
.	Windscreen Wiper	942-58116

5.2 Head Light & Cab Main Light (942-58112)

The operator can operate headlights of occupied cab on the train set, also control high/low beam of the headlight by using the Head Light switch (HLS).

5.2.1 Head Lights "On":

Headlights of the occupied cab will be operational in any of the following conditions:

- On availability of the auxiliary power (after pressing and releasing the AUX ON), headlights of the leading driving cab can be turned on by selecting low or high position of Head Light Switch (HLS).
- At the time of Call-On function, headlights can be turned ON by selecting low or high position using HLS.

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5.2.2 High/Low Beam Control (Head Lights)

Head Light Switch (HLS) toggles headlight's beams between "High Beam", "Low Beam" and "OFF" position.

5.2.3 Cab Main Light (CML)

Cab Main light installed on the driving cab ceiling would turn on automatically, in any of the following conditions:

- Pressing the Cab main Light Push Button (CMLPB1 or 2).
- Opening the Cab side Door (CSDOS(L) / CSDOS(R))
- Opening the Cab Saloon Partition Door (CSPDS1 or 2) on the un-occupied cab. In this condition, only Cab Main Light of un-occupied cab will turn on. This will also cause a visual alarm in TCMS VDU, in the leading DMC, triggered by TCMS.



NOTE

In spite of the train shutdown the cab main light will not be turned off automatically, without manual switching off. When leaving the train after shutdown, manually switch off the cab main light.

5.3 Tail Light, FL& Console Light (942-58113)

5.3.1 Tail Lights "ON" (Head Lights "OFF")

Taillights on the DMC car of the train set will turn on, in any of the following conditions:

- When the train system is powered up (i.e. AUX ON button is pressed), taillights on rear DMC car of the train set is turned on in the color of red.
- When the train system is powered up (i.e. AUX ON button is pressed), taillights on leading DMC car of the train set is turned on in the color of white. (Marker Light)

5.3.2 Flasher Light (FL)

Flasher light would turn on automatically, in any of the following conditions:

- Pressing the Flasher Light Manual Push Button (FLMPB)
- In case Safety Cut out Switch (SCS) is turned to "Cut out" position, the flasher light (FL) on both DMC cars are automatically turned ON in order to attract the attention of the driver of the following train or a train approaching from the opposite direction.
- In case Mode Selector is placed in Forward / Reverse position, the flasher light (FL) on both DMC cars are automatically turned ON.

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5.3.3 Driving Console Light (DCL)

Driving Console Light (DCL) installed on both sides of driver's cab can be switched on by using the Driving Console Light Push Button (DCLPB(R, L)).

5.3.4 Gauge Light

Gauge Light for Brake Cylinder (BC) / Main Reservoir (MR) / Brake Pipe (BP) and Parking Brake (PB) on the both cabs will turn ON when auxiliary supply is available.

5.4 Lights (942-58114 & 942-58115)

All saloon lights are DC 110V supplied LED saloon lights. To avoid simultaneous failure of all saloon lights, the power is supplied via four MCBs, ELCB1, ELCB2, ELCB3 and ELCB4.

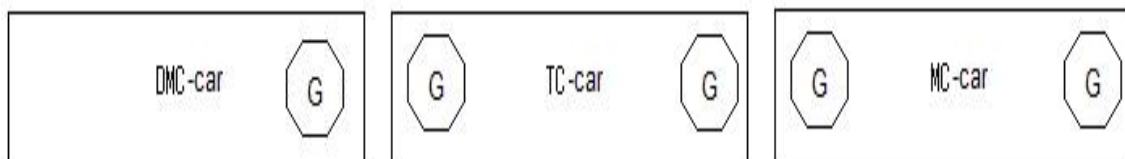
5.4.1 Saloon Light

All Saloon LED lights are automatically turned on when power is supplied for the lights. Light sensors inside the vehicle measure the light levels and the Auto dimmer unit adjusts the light when light levels are higher than the required levels.

In case of Auto dimmer failure, there is a safety mode in Saloon lights. If the saloon light signal is lost, which is normally coming from Auto dimmer, the saloon light will set maximum illumination.

5.4.2 Gangway Light

Gangway light is DC110V powered light and it draws power from the battery.



Where, G : Gangway Light

5.4.3 Windscreen Wiper (942-58116)

The windscreen wiper and washer system is provided for wiping and washing the cab front windscreens.

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6. Coupling & Uncoupling

6.1 Introduction

In this section, circuit descriptions for coupler control are provided.

Description	Dwg No
Coupling & Uncoupling Control	942-58117

6.2 Coupling & Uncoupling on the Driving Cab (942-58117)

6.2.1 Preparations for Coupling

The defective and rescue train set should be checked by permissible personnel first, before actual coupling is carried out;

- Place the two trains on the straight railroad
- Make sure that the defective train is safely parked and that all parking brakes are applied in the defective train.
- Make sure that status of the defective train set-whether it is available to couple with the mechanical coupler by train operator.
- In the event that mechanical coupling is available, press the COLPB (Call On Light Push Button) at the operating panel no. 3 at the leading cab back wall.
 - The tail light will be flashed in the both cabs of defective train.
 - The flashing of tail light in the defective train means that its status is available to couple mechanically.

6.2.2 Coupling

- Slowly move the train toward the defective train and stop the train before 1 m of it. Slightly move the train under the speed of 5 km/h.
- After coupling, tail light of defective train would be kept with flashing when the coupled train is approaching to the station, the staff in station knows that whether the approaching train is rescued or not and what is the coupled status of the defective train.
- When the train is approaching to the station, the status of the defective train can be distinguished according to the status of tail light.

6.2.3 Uncoupling

- Pressing and releasing of Uncoupling Pushbutton (UCPB) at the operation panel no.2 with few seconds in the normal train set
- Energizing of Mechanical Coupler uncoupled magnetic Valve (MCUCMV)
- Uncoupling of Mechanical coupler
- Return to the original position of Inductive Sensors (SS1 and SS2) and de-

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energizing of MCDR

(e) Move the healthy train set to reverse direction

(f) After power-up on the defective train, pressing and releasing of Uncoupling Pushbutton (UCPB) with few seconds will retract coupler head.

6.3 Functions

6.3.1 Mechanical Coupler Detection Relay (MCDR)

MCDR is energized when the inductive sensors are sensed.

6.3.2 Uncoupling Pushbutton (UCPB)

The push button is for uncoupling of the front automatic coupler (cab end).

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7. Traction System

7.1 Introduction

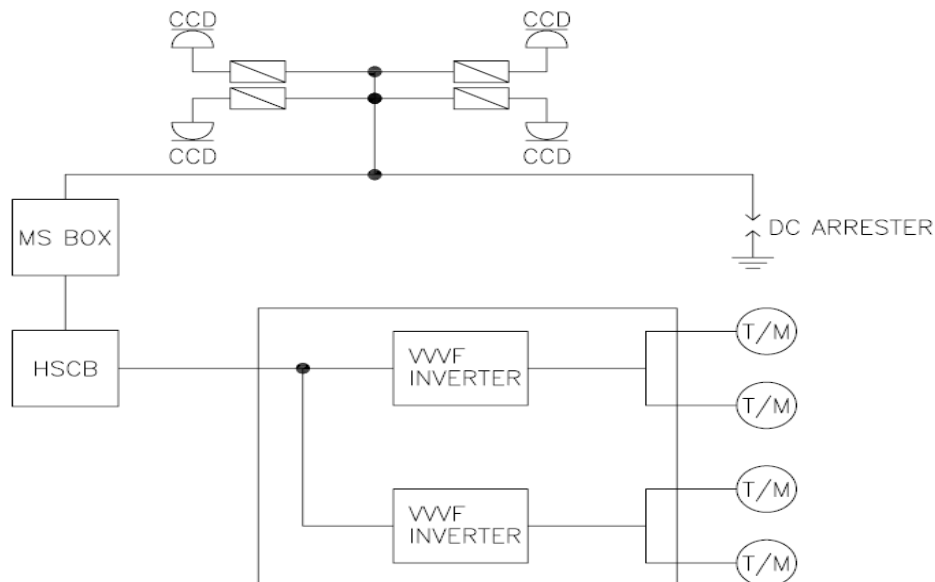
In this section, circuit description for DC750V main power supply, speed control, VVVF Inverter interface and ground are described.

Description	Dwg No
Main Power Supply	H7L7956
Speed Control	942-58119
VVVF Control	942-58120
Traction Return Current	942-58121

7.2 Main Power Supply (H7L7956)

7.2.1 Main Power Circuit

DC power, 750V, is supplied from Third rail current collector through the Main switch Box and High Speed Circuit Breaker (HSCB) to the VVVF Inverter. The DC arrester protects circuit against the external and internal lightning surge by current collector device.



VVVF inverter works on VVVF (Variable Voltage Variable Frequency) principle to control the rotation of the traction motor, i.e., the speed of train. It can switch powering/regenerative operation and forward/reverse operation, without interrupting the main circuit, by slip frequency and output phase permutation controls, i.e., only by controlling gate signals.

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7.3 Speed Control (942-58119)

The train operation mode can be determined by operation of Mode Selector (MS) and Traction Brake Controller (TBC) arranged on the driver's desk.

The position and corresponding functions of the TBC and MS are as follows:

7.3.1 Train Speed Control

7.3.1.1 Train Operation Mode Train Line Function

The train operation mode can be determined by operation of Mode Selector (MS) and Traction Brake Controller (TBC) arranged on the driver's desk.

The position and corresponding functions of the TBC and MS are as follows:

(a) OFF

MS of non-occupied driving cab has this position. In this mode of the leading cab, traction system is inoperative and full service brake is applied. Driver can move MS to "OFF" position and remove the master key. Mode selector cannot be moved from OFF, without occupation of key.

(b) Reverse Mode (REV)

In this mode, "reverse" signal from MS is transmitted to VVVF inverter of each motor car and traction and braking respond to Master Controller (TBC), in the reverse direction.

Mode selector can be moved to any position if key is ON and Master controller handle is at "Bmax" position.

(c) Wash/Coupling Mode (WCM).

For train wash and coupling, speed forward up to 3km/h. Mode Selector can be moved to any position if key is ON and Master controller handle is at "Bmax" position.

(d) Standby(STANDBY)

In this mode, the Master Control key is removable and the motoring is not capable by the traction inverter. Master Controller handle cannot be moved from "OFF" if key is OFF.

(e) Restricted Manual Mode(RMM)

Manual control backed up by Propulsion logic ATP (max speed 25 km/h). Mode Selector can be moved to any position if key is ON and Master controller handle is at "Bmax" position.

(f) ATP Mode (ATP)

Manual control backed up by ATP. Mode Selector can be moved to any position if key is ON and Master controller handle is at "Bmax" position.

(g) ATO Mode (ATO)

Automatic train operation backed up by ATP. Mode Selector can be moved to any position if key is ON and Master controller handle is at "Bmax" position.

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When the powering train-line command comes from train operator, these are transmitted to propulsion system via train lines. This initiates the train movement by propulsion system, upon receiving powering signals with PWM signal from the controlling cab.

The powering train-line would be de-energized and train cannot move at the station, even if TBC is set to "Powering" position, in any of following conditions;

- Door Proving Relay is dropped (all doors of train set are not closed and locked)
- Full Service Brake is applied by ATP
- Drivers Brake Valve is operated by operator

In the event that the failure of DPR or "FSB" of ATP, these signals can be overridden by turning the DPLCOS to "Cut out" position or the operation of SCSR, which is energized by the SCS switch in "Cut out" position.

The operation of drivers valve (brake handle), installed on the driver's cab, will energize BPPSR relay and cut off the powering signals to prevent damage of the wheels.

After selection of TBC to "Brake or Emer. Brake" by train operator under the MS to "REV/WCM/STANDBY/RMM/ATP" position, train will be braked by manually.

In case of ATO mode, ATO will apply train brake automatically.

Braking command signal would also be generated in any of the following conditions;

7.3.1.1 Function

The response of sub-systems, corresponding to powering and braking commands, is shown below;

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	State1	State2	State3	State4	State5	State6	State7	State8
Powering (Trainline)	0	0	0	0	1	1	1	1
Braking (Trainline)	0	0	1	1	0	0	1	1
PWM Trainline / TMS brake demand	0	1	0	1	0	1	0	1
GCU (VVVF)	C	C	B _{max}	SB	M50	M	C	C
BCU	—	—	B _{max}	SB	—	—	—	—
Failure contact			Major Event					

“0” = Invalid signal

“1” = Valid signal

“—” = No braking effort provided by BECU

“B” = Braking according to value of PWM-signal

“B_{max}” = Max. service brake application at tare load condition

“SB” = Service brake depending on PWM brake demand

“M” = Powering according to PWM-signal

“M50” = 50% of maximum torque at each load condition

“C” = Coasting

PWM Signal

PWM output signal is based on the powering or braking effort and can be delivered to control circuit of VVVF Inverter and BECUs at each car because signal information is pulse width-not analogue voltage and pulse width information is carried by amplified voltage with high reliability.

In case of Manual operation, VVVF Inverter, BECU and TCMS receive the powering/braking PWM signal directly at the leading cab from the potentiometer of the Traction Brake Controller. But, in case of ATO operation, VVVF Inverter, BECU and TCMS receive the powering/braking and PWM signal at the leading cab from the ATC through RS485 communication.

7.4 VVVF Inverter Interface (942-58120)

The VVVF Inverter control power (DC) is supplied after pressing and releasing of "AUX ON" button and shutdown after pressing and releasing of the "AOFFS" button.

Input signal of VVVF Inverter is as follows;

- Forward signal
- Reverse signal
- SCS cut out signal
- MS "ATP" signal
- MS "RMM" signal
- MS "WCM" signal
- Powering 1 and Powering 2 signal
- Braking 1 and Braking 2 signal
- PWM signal
- Emergency Brake command

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- HSCB status signal
- Speed signals from speed sensors

Output signal of VVVF Inverter is as follows;

- VVVF Inverter fault signal to TCMS & fault indicator

Interface signal between VVVF Inverter and BECU is as follows;

- Load weight value
- ED Cut-off signal
- ED feedback value
- ED Brake active
- Request Holding Brake
- Sliding

7.5 Grounding (942-58121)

To minimize the effects of Electro Magnetic interference in the earth circuits the following measures are taken:

- The negative line of VVVF Inverter, which feeds the variable voltage and frequency to Traction Motor, is connected to Earth Brush via Earth Terminal Box.
- To minimize high frequency circulating currents, due to the capacitance between the motor stator coils and frame, the earth line of traction motor is twisted with motor leads and is terminated at the single earth point located inside of the VVVF Inverter equipment box.
- The low tension lines, such as the negative line of APS, the negative line of all loads which are fed by APS and Battery negative line are connected to Earth Brush via Low tension Earth Terminal Box (LETB) different from that of high tension.
- Low tension DC negative return cables within a unit set are connected to the low-tension earth bar on the car to which the battery negative is connected.
- The equipment housing for AC 415V & AC 230V is connected to the low-tension earth bar.

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8. Brake System

8.1 Introduction

In this section, Compressor Motor control, Safety Proving Brake Loop, Brake Control, etc. is described.

Description	Dwg No
Compressor Control	942-58123
Brake Loop	942-58124
Emergency Brake	942-58125
Parking Brake	942-58126
Horn Control	942-58127
Brake Control Circuit, (DMC, MC)	942-58128
Brake Control Circuit, (TC)	942-58129

8.2 Compressor Control (942-58123)

415V AC, 50Hz motor, powers the compressor. The main air-compressor unit is controlled by one relay contact "R04" of BECU, which is controlled by software to 8.0 bar for cut in and 10 bar for cut out. The second pressure switch (Compressor Motor Governor (CMG)), which can be set to 7.5 bar for cut in and 10 bar for cut out is provided as hardwired back-up in case of failure of BECU.

8.2.1 Treatment of Air Dryer Unit (ADU)

When the compressor operates, the air dryer system treats compressed air. The starting signal of air dryer unit is incorporated with CMC contact of EDB. TCMS will monitor the status of Air dryer unit (ADU) evaluated by Air Dryer Pressure Switches (ADPS1 & 2).

8.2.2 Record of Compressor Running Time

The total running time of compressor is recorded by BECU, whenever the compressor - operating signal, CMC, is given by EDB to BECU.

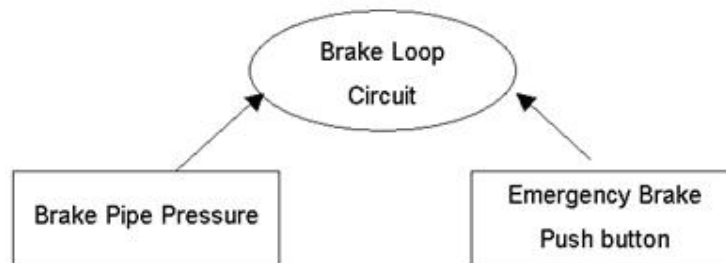
8.2.3 Operation of Automatic Drain Magnetic Valve (ADMV)

The Auto Drain Magnetic Valve (ADMV) is used in compressed air systems for automatic drainage of condensates from compressed air reservoir, oil/water separator and air dryer unit. As the operation of ADMV is related with working of compressor motor, it works whenever the compressor motor is working.

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8.3 Brake Loop (942-58124)

The brake loop circuit monitors the various safety related factors, for operation of brake, through the train set. Any abnormality such as operation of emergency brake push button, Brake pipe pressure, etc will cause interruption of the normal brake loop, which will cause penalty application of the emergency brakes.



The brake loop control circuit is a dedicated hardwired circuit that controls the emergency brake application.

Following conditions should be fulfilled to complete the brake loop circuit;

In the leading DMC,

- HCR of leading cab is to be energized.
- Both EBPB1 and EBPB2 are not operated.
- Brake Pipe Pressure is above 4.8bar and not drop below 3.5 bar.

In the TC,

- MRPS Pressure is above 7.0 bar and not drop below 6.0 bar.

In the MC,

- Brake Pipe Pressure is above 4.8bar and not drop below 3.5 bar.

In the rear DMC,

- MS(Mode Selector) is "OFF" condition.
- Both EBPB1 and EBPB2 are not operated.
- Brake Pipe Pressure is above 4.8bar and not drop below 3.5 bar.

The completion of brake loop on the normal train set energizes BLPR relay. As a result, train operator on the leading cab can apply service brake and emergency brake (by completion of emergency brake loop by BLPR contact).

In the event of normal brake loop failure, the selection of BLCOS maintains the emergency brake loop.

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8.3.1 Function

8.3.1.1 Brake Loop Proving Relays (BLPR)

The relay is energized when brake loop is completed. The de-energizing of BLPR applies emergency brake to the train set.

8.4 Emergency Brake Loop (942-58125)

This circuit describes the control of Emergency Brake Magnet Valve (EBMV). To complete the circuit, Brake Loop circuit should be setup.

The two emergency brake loop circuits monitor the various safety related factors for operation of emergency brake, through the train set. Any abnormality such as choice of brake loop, operation of emergency brake push button, position of master controller (TBC) and dead man safety device, etc will cause application of emergency brake.

There are two emergency brake-monitoring circuits on the leading DMC car, one is for normal and the other is for redundancy. The selection of the emergency brake circuits is determined by the operation of Emergency Brake Loop Selection Switch (EBSS).

The Emergency Brake Loop Relay1 (EBLR1), on the normal emergency brake-monitoring circuit would be energized when all of the following conditions are achieved;

- HCR to be energized.
- Emergency Brake loop Selector Switch (EBSS), in the activated driving cab is to be at "Normal".
- The train is at standstill. The required relay interlock is provided by ZVAR, which is driven by the ATP system.
- EBPB1, 2, Emergency Push button switch, in the driving cab is not to be operated.
- Master Controller (TBC) should not be in EB position.
- The emergency brake auxiliary A relay (EBAR), which is driven by the ATP system are energized.
- Pressure of Main Reservoir provided in TC-car is more than 7.0bar.
- When driving cab is active, inhibited mode changes should not take place.
- The brake loop is completed.
- The driver holds Dead Man Safety Device (DSD) – DMSR1.
- EBLR1 is energized.

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The Emergency Brake Loop Relay2 (EBLR2), on the redundant emergency brake-monitoring circuit would be energized when all of the following conditions are achieved;

- HCR is energized.
- Emergency Brake loop Selector Switch (EBSS) in the activated driving cab is to be at "Redundant".
- The train is at standstill. The required relay interlock is provided by ZVAR, which is driven by the ATP system.
- EBPB1, 2, Emergency Push button switch, in the driving cab is not to be operated
- Master Controller (TBC) is not EB position.
- The emergency brake auxiliary A relay (EBAR), which are driven by the ATP system are energized.
- Pressure of Main Reservoir provided in DMC-car is more than 7.0bar
- When driving cab is active, inhibited mode changes should not take place.
- The brake loop is completed.
- The driver hold dead man safety device (DSD) – DMSR2
- EBLR2 is energized.

8.4.0 Activation of Dead-man Safety Device Relay (DMSR)

The Dead Man Device (DSD) is a safety device to keep train operator alert and attentive. The DSD is available on master controller (TBC) and the operator should keep it continuously twisted to avoid emergency brakes. The twisting of DSD is monitored by Dead Man Safety Device Relay (DMSR) 1 & 2.

8.4.1 Emergency Brake Loop Circuits

The emergency brake loop circuit monitors the various safety related factors for operation of emergency brake, through the train set. Any abnormality such as choice of brake loop operation of emergency brake push button, position of traction brake controller and dead man safety device, etc will cause application of emergency brake.

In the event of the failure of emergency brake normal loop circuit-1, the emergency brake can be released by using redundant emergency brake monitoring circuit 2.

8.4.2 Emergency Brake Loop 1

The Emergency Brake Valve Relay (EBVR) 1s on the each car will be energized in accordance with the completion of emergency brake loop 1.

8.4.3 Emergency Brake Loop 2

The Emergency Brake Valve Relay (EBVR) 2s on each car will be energized in accordance with the completion of emergency brake loop 2.

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8.4.5 Energizing of Emergency Brake Magnetic Valve (EBMV)

The Emergency Brake Magnetic Valve (EBMV) on the BCU of each car will always be energized in accordance with the completion of brake loop.

In case EBMV is de-energized, the emergency brake will be applied.

8.4.6 Function

8.4.6.1 Brake Loop Cut Out Switch (BLCOS)

In the event of normal brake loop failure, BLCOS is to be operated to override interlocking function of the Brake Loop circuit and control directly the emergency brake.

The BLCOS has two positions as below:

- **NORMAL:** default position. In this position, the normal service brake by VVVF Inverter and BECU can be applied.
- **CUTOUT:** In the event of the failure of normal brake loop or service brake, the train set will be applied with emergency brake immediately. To avoid application of emergency brake on train set, BLCOS is to be operated at "CUTOUT" position to override interlocking function of the Brake Loop circuit, with mode selector (MS) position (TBC) to "Forward", and control directly the emergency brake.

8.4.6.2 Emergency Brake Loop Selection Switch (EBSS)

The EBSS has two positions as below:

- **NORMAL:** It's a default position. In this position, the interruption of normal emergency brake loop results in application of emergency brake.
- **REDUNDANT:** In case of normal emergency brake loop failure, the emergency brake of the train can be released by operating EBSS to "REDUNDANT". Selecting of "REDUNDANT" changes the emergency brake loop used from "NORMAL" to "REDUNDANT".

8.4.6.3 Safety Cut-out Switch (SCS)

When the SCS is selected to "Cut-out" during ATP system failure, the emergency brake loop circuit is completed via contacts of SCS.

8.4.6.4 DSD Bypass Selector Switch (DSDBS)

When the DSDBS selector switch is selected to "Bypass" during Master Controller DSD contacts malfunction, the emergency brake loop circuit is completed via contacts of DSDBS.

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8.5 Parking Brake (942-58126)

When the train is at standstill, once the PBAPB (Parking brake Applied push button) is pressed from the controlling cab, the parking brakes of the train are applied by Impulse Magnet Valve (PBMV) until PBRPB (Parking Brake Released Push button) is pressed.

8.5.1 Functions

8.5.1.1 Parking Brake Pressure Switch (PBPS1 & 2) & Parking Brake Relay (PBR)

Parking Brake Pressure Switch (PBPS) monitors the air pressure for parking brake. When parking brake is released, PBPS of all cars connected and Parking Brake Relay (PBR) is energized.

8.6 Horn (942-58127)

When Horn Push Button is pressed on either cab, horns of that car apply both tones on succession until the release of horn pushbutton.

8.7 Brake Control (942-58128 & 942-58129))

This sheet is described about interface I/O signal with BECU.

- ☐ Power supply
- ☐ Emergency Brake
- ☐ Braking Command
- ☐ Car Selection
- ☐ Brake Demand PWM Signal
- ☐ ED Brake Active
- ☐ ED Brake Cut-out
- ☐ Request Holding Brake
- ☐ ATO Cut-out
- ☐ Apply Brake ATO Cut-out
- ☐ Load Signal
- ☐ WSP Protection
- ☐ Speed Signal
- ☐ CV-Pressure Signal
- ☐ RS485 Communication between BECU and TCMS
- ☐ BECU Event Log to TCMS
- ☐ Main Air Compressor Motor Signal

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9. Auxiliary Power System

9.1 Introduction

In this document, circuit for APS control, AC 415V shore supply and 110V power supply are described.

Description	Dwg No
APS	942-58130
AC 415V Shore supply	942-58131
Battery Control	942-58132

9.2 APS (942-58130)

9.2.1 APS Control

The auxiliary power supply will supply the stable 3 phase AC power to air-conditioners, AC Compressor, 230VAC sockets and control circuit etc.

The power to the auxiliary power supply will be supplied from the third rail.

The output of the Auxiliary Power Supply is 415VAC, 230VAC and 110VDC.

The APS control power (DC) is supplied as soon as STUR is energized and shutdown in case of APS start signal is cut-off.

Input signal of APS is as follows;

- APS Start(Auxiliary On)
- APS Reset

Output signal of APS is as follows;

- APS fault signal to the indicator of driver's desk
- Healthy signal of APS
- Power Extension On

9.2.1 Input Circuit

750VDC power will be supplied from the third rail.

9.2.2 Battery Charger Circuit

Battery Charger circuit will supply the DC power to battery and control power etc.

The power to the Battery Charger will be supplied from 750VDC power of the third rail.

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The output voltage of the Battery charger is 110VDC and it will be stabilized in constant under input voltage variations and output load variations.

9.2.3 230VAC Single Phase output

230VAC single phase output voltage is transformed from AC output voltage of the auxiliary power supply with the transformer.

The APS unit distributes AC 230V single phase supply to the following,

- Each car is equipped with 230VAC socket outlets to enable the use of vacuum cleaners.
- Each car Relay/EDB panel is equipped with two Fans for air circulation & Ventilation.
- Also, each car is equipped with 230VAC sockets for Passenger Laptop and Mobile Charging.

9.2.4 415VAC 3 Phase output

The APS unit supplies 415V AC, 3 phase to cab and saloon air-conditioner units, Air compressor.

9.3 Shore Supply, 415VAC (942-58131)

9.3.1 Shore Supply Unit Control

In case of maintenance of Air conditioners without 750VDC line input, the power of 3 phase 415VAC is supplied via Shore Supply Box to load for maintenance.

Two Shore supply box are equipped in TC car. 415VAC power is connected to one of the Shore Supply Box.

If 415VAC power is connected to one of the shore supply box and following conditions for the closing SSK1 is satisfactory, the contactor SSK1 is closed. When 415VAC power line is energized, the lamp SSL is lighted.

If APS detects that voltage is supplied to 415VAC line, without operation of APS, APS energize the coil of the contactor ESK equipped in Power Extension Box and contactor ESK is closed. Then Shore Supply Power is supplied to 3 cars.

The actual usage of the shore supply unit is as follows;

- 230VAC single phase sockets
- 415VAC three phase for battery charging
- 415VAC three phase for one VAC unit to test at a time

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9.4 Power Extension Box (942-58130)

In case of an Auxiliary power supply's failure, the power of three-phase AC415V is supplied from another Auxiliary Power supply in the same car via the Power Extension Circuit.

9.4.1 Power Extension Control Circuit

Fault of the Auxiliary Power supply is detected by another Auxiliary Power Supply in the same car. Another Auxiliary Power Supply energizes the power relay in the Power Extension Box. Then the contact of the power relay is closed and the contactor for the power extension(ESK) is energized.

The circuit for energizing the contactor for the power extension (ESK) has interlock circuit, which is consisted of two relays (VDR1 and VDR2). These relays watch the each side's bus line voltage of ESK in order to prevent ESK from being closed in case that both sides of ESK are energized at the same time.

The condition of ESK is monitored by TCMS.

9.5 Battery Control (DC110V Power Supply) (942-58132)

9.5.1 Power via "D2" Line

The DC 110V via "D2" line comes from battery directly. This power is used for the following functions;

- Controlling cab circuit 1
- Start-up relay circuit
- Train line supply contactor 1
- Cab main light
- Tail light and Flasher light
- Horn
- Shore supply circuit

The "D2" line is protected by battery fuse against short circuit and it has separate MCB for each load, before the load or equipment.

9.5.2 Power via "D1" Line

The DC 110V via "D1" line comes from BIC (Battery Isolation Contactor). This power is used for the normal loads except above loads for "D2" line and will be available after train start-up.

- Feeding with DC power via "D1" line

After closing of BIC, DC power can feed normal loads with DC 110V via "D1" line. The "D1" has separate MCB for each load against short circuit, before the load or equipment.

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9.5.3 Back up of DC Power

In the event of APS failure, there is no AC power and DC charging power available from APS.

But, the BIC contactor is still closed by D2 line and hence "D1" line is also available. The battery installed on the TC car can feed DC power via 'D1" line to train set for 1 hour after failure of APS.

The BIC remains energized until train shut down or detection of low battery voltage by BVMR.

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10. Door System

10.1 Introduction

In this section, Saloon Door Supply Voltage, Door Operation Left/Right, Door Proving Loop, Local Door Interlock, Door Communication with TCMS is described.

Description	Dwg No
Saloon door supply voltage	942-58137
Door operation, Left	942-58138
Door operation, Right	942-58139
Door proving loop	942-58140
Local door interlock	942-58141
Door communication with TCMS	942-58142

The Door Operation can be controlled in two modes. The Manual Mode and Auto Mode operation shall be selected through sealed 'Door Mode' Selector Switch in driver CAB. Below figure shows the Door Mode selector switch.



Figure 19. Door Mode selector switch - Driver cab

1) Door Open Operation:

Door Mode Switch in Manual Mode

MS in Standby Mode: In this mode, doors will open train in standstill condition only by pressing 'Door Open Push button' on CAB desk Operating Panel.

MS in ATP Mode: In this mode of operation, based on platform side train operator will press the Door Open Push Button on operating panel and accordingly signaling CC will receive the Door open command and CC gives the Door open command to each Door Control Units (DCU) through the train line for correct side door opening.

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Door Mode Switch in Auto Mode

MS in Auto Mode: In this mode of operation, Doors will automatically open at correct platform side by signaling CC. CC gives the Door open command to each Door Control Units (DCU) through the train line for correct side door opening without train operator intervention.

2) Door Close Operation:

Door Mode Switch in Manual / Auto Mode

MS in ATP / ATO Mode: In this mode of operation, based on platform side train operator will press the Door Close Push Button on operating panel and accordingly signaling CC will receive the Door Close command and CC gives the Door Close command to each Door Control Units (DCU) through the train line for door closing. When all doors are closed, the "All Door Closed" indicator on the driver's cab would be illuminated.

MS in Standby Mode: In this mode of operation, on pressing and releasing of the door close pushbutton on the driver's cab, a door close signal would transmit to the individual Door Control Units (DCU) through the train line. When all doors are closed, the "All Door Closed" indicator on the driver's cab would be illuminated.

3) Obstacle detection during Door Closing Operation:

Door Mode Switch in Manual Mode

MS in Standby Mode: In the event of obstacle detection during Door Closing, the particular door will attempt to re-close 3 times. if obstacle still persists after 3 attempts, the door will go to and remain in full open position. further door closure shall require another activation of the corresponding "Door Close" command from train operator through Door Close Push button.

MS in ATP Mode: In the event of obstacle detection during Door Closing, the particular door will attempt to re-close 3 times. if obstacle still persists after 3 attempts, the door will go to and remain in full open position. As per Signaling contractor, Train operator will press the Door Open Push Button (signaling CC will give open command and all doors will open) and press the Door Close Push Button on operating panel and accordingly signaling CC will receive the Door Close command and CC gives the Door Close command to Door Control Units (DCU) through the train line for door closing.

Door Mode Switch in Auto Mode

MS in ATO Mode: In the event of obstacle detection during Door Closing, the particular door will attempt to re-close 3 times. if obstacle still persists after 3 attempts, the door will go to and remain in full open position. As per Signaling contractor, Train operator will change the DMS to Manual mode and press the Door Open Push Button (Signaling CC will give Open command and all doors will open) & press the Door Close

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Push Button on operating panel and accordingly signaling CC will receive the Door Close command and CC gives the Door Close command to Door Control Units (DCU) through the train line for door closing.

Please refer Door control circuits drawings 942-58138D & 942-58139D for more details.

TMS will monitor circuit integrity of door control circuit, against any abnormality and display the fault (if any) on the VDU of driver's desk. The block diagram for door opening/closing interface shown in below figure.

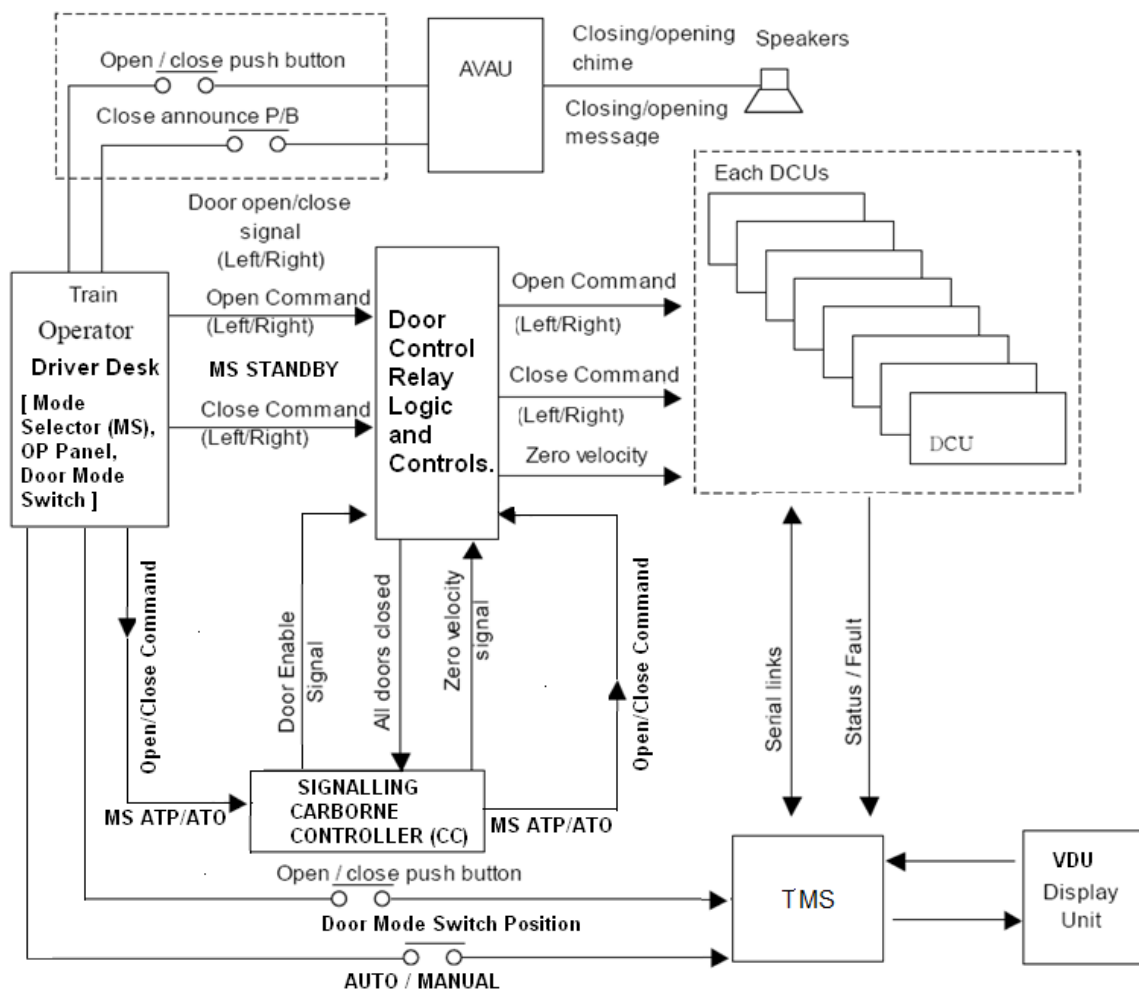


Fig. 2A-09-00-00-02 Man-Machine Communication

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10.2 Saloon Door Supply Voltage (942-58137)

Saloon doors of the train set are supplied with 110VDC via separate MCB's in each side of the car respectively.

DC power on each EDCU can be isolated using the corresponding Door Isolation Switch (DIS). This enables opening of the doors manually.

10.3 Saloon Door Operation (942-58138 & 942-58139)

In this section, opening/closing operation of the saloon door is described.

10.3.1 Zero Velocity Signal

When the train is at standstill, a Zero Speed Signal is transmitted to all train doors via train line. This makes the door control circuit operative, upon receiving door open command from the controlling cab.

10.3.2 Saloon Door Opening

The opening operation of the passenger saloon door would take place under following condition:

- A door opening command can be generated from the occupied cab only, upon receiving a door enable signal from ATP. Signal reception from ATP can be verified by the operation of Door Open Enable Left/Right Indicator (DOELI, DOERI) incorporated with two door open push button in the driver's panel, in operating cab.

In order to avoid incorrect operation, an interlock is provided between contacts of door close pushbutton and door open pushbutton.

10.3.3 Saloon Door Closing - Left

The closing operation of the passenger saloon door would take place under following condition:

- A door closing command is generated from the occupied cab. In order to avoid incorrect operation, an interlock is provided between Contacts of door close pushbutton and door open pushbutton.

The following table summarizes the various combinations of open/close logic performed by EDCU:

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Combination of Open/Close Logic for door

Zero Speed	Close Command	Open Command	Previous Door State	Response of DCU
0	X	X	Clo	Maintain Previous
			Clo	Maintain Previous
1	0	0	Op	Maintain Previous
			Clo	state
1	0	1	Op	Maintain Previous
			Clo	Open Door
1	1	0	Op	Close Door
			Clo	Maintain Previous
1	1	1	Op	Close Door
			Clo	Maintain Previous
DC 110V of DCU “ON” or “OFF”			Op	Maintain Previous
			Clo	state

"1" means the command is valid.

"0" means command is invalid.

"X" means don't care state.

10.4 Door Proving Loop (942-58140)

The door proving loop monitors the local door interlock relays through the train set. Any abnormality of local door interlocking will cause interruption of all doors close loop, which will cause interruption of Traction circuit.

Once DPR (coil) is energized, it holds its status by DPR contact till opening of saloon door.

10.4.1 Function

10.4.1.1 All Door Closed Relay (ADCR)

All doors close relay (ADCR) is energized by the activation of local door interlock summary relays, LDISRR and LDISLR, on successful closure of the doors on each side of the car.

When the ADCR is energized, all doors close lamp (ADCLp) will be lit.

10.4.1.2 Door Proving Relay (DPR)

The closure of all doors and energizing of ADCR causes to energize Door Proving Relay (DPR). However, in case ADCR is de-energized, due to some failure in the door loop, during motion, the traction would still be available through interlocking function of Door proving relay (DPR), until a door opening command is requested, at the next station.

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In the event of non-availability of traction circuit by failure of DPR, all doors close line being healthy, DPR can be bypassed by moving Door Proving Loop Cut Out Switch (DPLCOS) in "bypass" position.

10.4.1.3 Door Proving Loop Cut Out Switch (DPLCOS)

In the event of normal door loop failure, DPLCOS is to be operated to override interlocking function of the Emergency Brake circuit.

The DPLCOS has two positions as below:

- **NORMAL:** default position. In this position, the normal door loop by local door interlocking relays is available.
- **CUTOUT:** In the event of failure of normal door loop, the train set will immediately apply emergency brake. To avoid application of emergency brake on train set, the selection DPLCOS to "CUTOUT" position is to be operated to override interlocking function of the normal door loop circuit.

10.5 Local Door Interlock Circuit (942-58141)

A Local Door Interlock Relays, LDISLR and LDISRR, on each side of each car, is energized based on individual door status for that side. The contacts of LDISLR and LDISRR are also incorporated with door proving loop as a part of safety circuit.

Operation (opening) of any local door causes the relays, LDISLR and LDISRR, to de-energize and causes Door Out side indication Lamp to illuminate, provided no passenger alarm is operated.

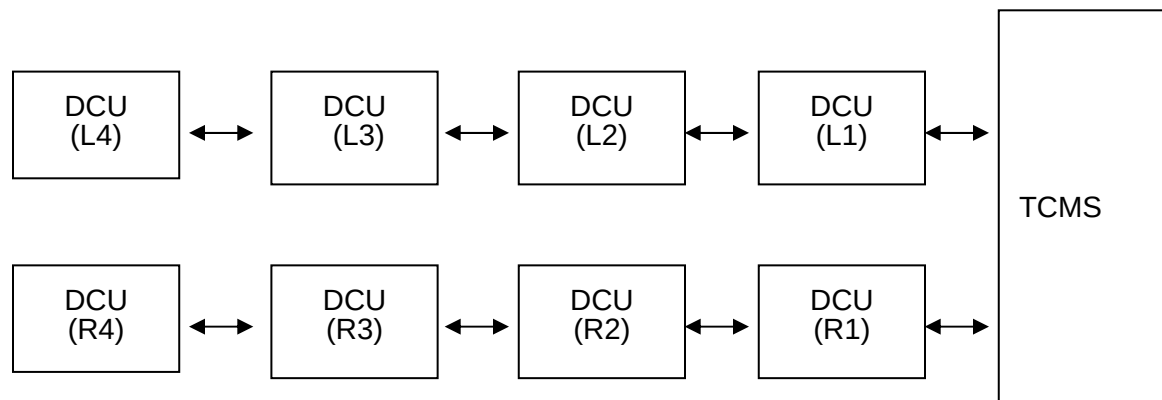
Operation of any passenger alarm handle however causes the door out side indicators to flash on both sides of the car.

In the event of a door unit failure(s), the door(s) can be isolated individually using Door Isolation Device.

10.6 Communication with TIMS (942-58142)

Door control units of each side of car will communicate with TIMS through RS485 serial data transmission for real time monitoring and troubleshooting.

For appropriate connection between TIMS and each DCU twisted screened cable is used.



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11. Air Conditioning System

11.1 Introduction

In this section, vehicle control circuit regarding air conditioning system is described.

:	Description	Dwg No
.	Cab VAC	942-58143
.	Saloon VAC Power	942-58144
.	Saloon VAC Control	942-58145

11.2 Cab VAC (942-58143)

The VAC unit operates on 3phase 415VAC, 50Hz provided via MCB by APS. The control circuit operates on 110VDC vehicle control power supply.

The cab air conditioning has control switch and an indicator (CVACFLp). The Cab VAC Control Switch (CABHVCS) is for main electrical control switch and has 6 positions as follows;

(a) Off

(b) Vent Low

(c) Vent High

(d) Auto

(e) Cool 1

(f) Cool 2

The Cab VAC Fault Lamp (CVACFLp) on the driver's desk reflects CAB VAC unit failure.

11.3 Saloon VAC (942-58144 & 942-58145)

The VAC unit operates on 3 phase 415VAC 50Hz provided via MCB by APS. The control circuit operates on 110VDC vehicle control power supply.

11.3.1 Address Bit Signal

To avoid inrush current during train initiation, saloon VAC units have staggered starting function. For this function, air conditioning unit 1 and 2 (both DMC, TC and MC car) would be differentiated by AD (Address Bit Relay) provided in each saloon air conditioning unit.

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11.3.2 Fault Indication

When VAC unit has failed, Saloon VAC Fault Lamp (SVACFLp) reflects the same in the driver's cab.

11.3.3 APS Status Signal

When 3 phase, 415VAC is available, Saloon VAC unit can be operated on the condition that a APS status signal is received.

11.3.4 Smoke Detection

When accidental smoke is sensed, an alarm signal is initiated at the smoke detector. These actions can facilitate the management of the toxic smoke and fire gases throughout the areas served by the air duct system. This signal will be transmitted to each VAC unit. The VAC unit would close the damper spring and circulate air inside.

11.3.5 Emergency Ventilation

In case 3-phase 415VAC is not available, an emergency Ventilation Fan is operated by converting 3-phase 415VAC power from 110VDC Battery power by emergency ventilation fan inverter in the VAC unit.

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12. Train Management System

12.1 Introduction

In this section, vehicle interface circuit regarding Train Management system is described.

Description	Dwg No
TCMS Interface	942-58146

12.2 General (942-58146)

TCMS is the complete system for controlling, monitoring, protecting the onboard systems and subsystem connected to the train communication network either directly or via discrete I/O.

12.2.1 Sub-systems on DMC

The Sub-systems linked and communicates with TCMS via Local Bus are followings.

- (a) Train Radio
- (b) ATO/ATP (CC) System
- (c) Display Controller
- (d) VVVF Inverter
- (e) BECU
- (f) VAC
- (g) EDCU
- (h) PA & PIS
- (i) CCTV

12.2.2 Sub-systems on TC/MC

The Sub-systems linked and communicates with TCMS via Local Bus are followings.

- (a) VAC
- (b) EDCU
- (c) BECU
- (d) APS

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13. Communication System

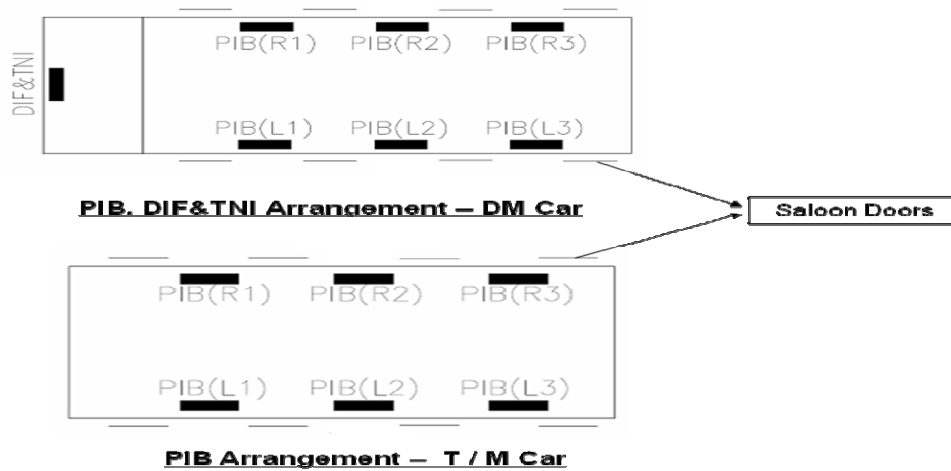
13.1 Introduction

In this section, vehicle interface circuit regarding Communication system is described.

Description	Dwg No
PIB & TII (DMC)	942-58147
PIB & TII (TC, MC)	942-58148
DVAS, PA System	942-58149
PA Amplifier (DMC, MC)	942-58150
PA Amplifier (TC)	942-58151
ATP Interface	942-58152
Train Radio Interface	942-58153
CCTV	942-58154

13.2 Passenger Information Display Screen (PIB) (942-58147 & 942-58148)

The LCD displays (PIBs) will be installed on side coving panel (flush mounted) of train for Passenger Information Display. All PIBs will indicate the animated graphic and next station etc information according to train running location.

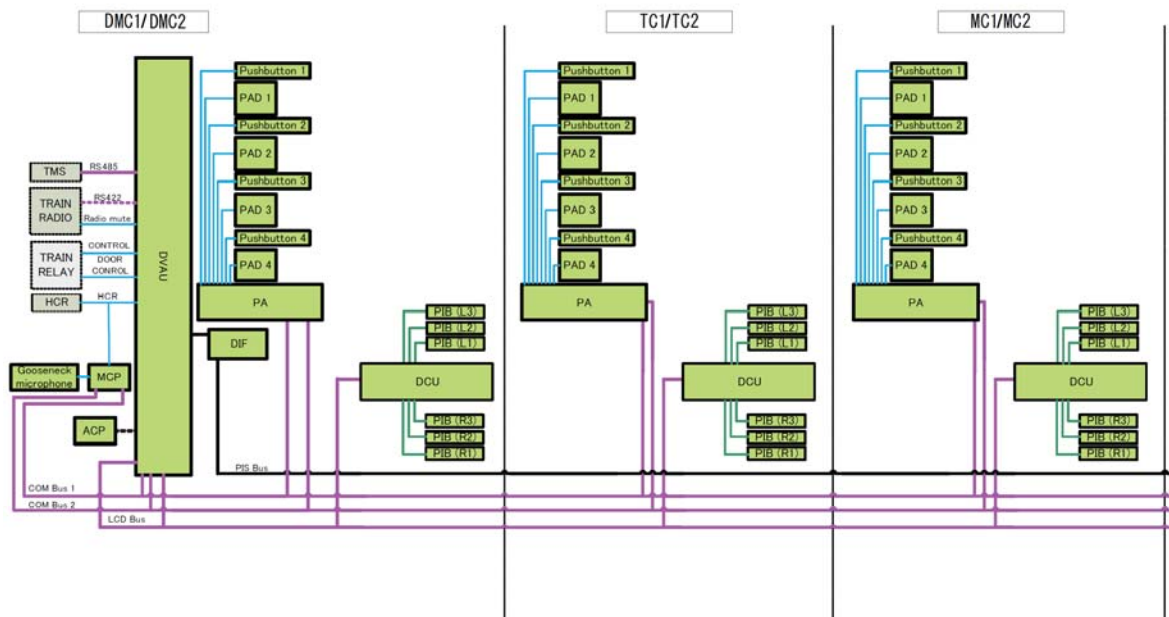


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13.3 Digital Voice Announcement Unit (DVAU) (942-58149)

The Digital Voice Announcement Unit (DVAU) is the main component of the Public Address (PA) system and controls the broadcasting of announcements.

13.4 Public Address System (PA) (942-58150 & 942-58151)



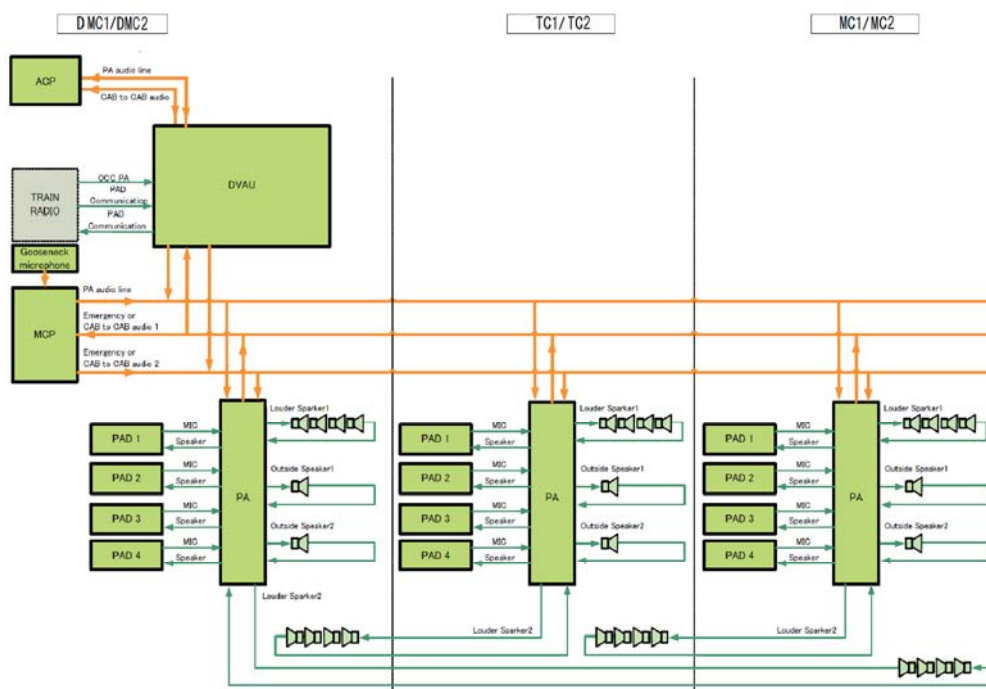
Configuration of PA-PIS (Representation Purpose only)

The major functions of PA system are given below:

- Pre-recorded Announcement (Automatic & Manual mode)
- Driver PA
- Cab to Cab Communication
- Emergency Communication (Passenger Alarm system)
- OCC/BCC PA (Radio Broadcasting by train traffic controller)
- Door Chime

13.5 Public Address Amplifier Unit (PAMP) (942-58150 & 942-58151)

The PA amplifies the signal from DVAU and broadcasts it to the loudspeakers in the passenger saloon. Power amplifier will be provided in each car to feed 50% of the speakers in the same car and 50% in the adjoining car, to ensure that in the event of a single power amplifier failure; at least half of the speakers are still operative in the car.



Configuration of PA Audio Line for speaker redundancy
(Representation Purpose only)

13.5.1 Passenger Emergency Alarm & Handle (PEA & PAEH)

Four PEA & PAEH units are provided in each car. The units are supplied by 24VDC power via PAMP. When a PEA is operated by manually by PAD, a visual indication on the exterior of the car (Door Indication Light) would flash to advise station staff

13.5.2 Passenger Information System

The PIS is an information system for the passengers. The passenger information system (PIS) shall comprise of the following:

- Electronic Information Display (EID) – 6 no.s of LCD/TFT displays for Passenger Information Display.
- Electronic Destination Display (EDD) – 1 no. Destination Indicator for Front (DIF) & Train Number Indicator (TNI) in the front of cab to indicate train number and destination station name.

The major functions of PIS are given below:

- Station Information display
- Special / Emergency message display
- Evacuation message display
- Advertisements

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13.6 ATP (CC) Interface

The Signaling contractor provides Automatic Train Protection (ATP) system.

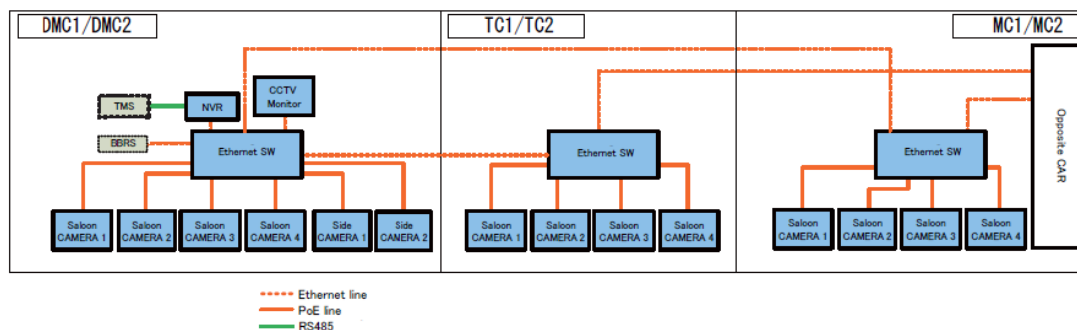
13.7 Train Radio Interface

The Telecommunication contractor provides Train Radio System.

13.8 CCTV

The Passenger Surveillance system comprise of a Close Circuit Television (CCTV) network using surveillance cameras, recorder and monitor for surveillance of saloon area and platform / door (station stops).

Backbone Ethernet runs through the complete train set and IP camera transmits compressed video images through the Ethernet and Network video recorder receive them to decompress and display on LCD touch monitor display installed on the driving cab. The digital video recorders also enables driver to search and playback of the stored video streams through LCD touch monitor operation.



Configuration of CCTV System
